ModelAudit — Summary (page 1)


```
{
  "dataset": "taiwan",
  "model": "scorecard",
  "scope": "research demonstration only; thresholds are local review heuristics",
  "evidence_sha256": "ea39a375cd9d45b17f654bb7b7531663fb184394ff5a1bf696b411eaf5e69865",
  "findings": [
    {
      "test_id": "STAB-CSI-01",
      "severity": "high",
      "confirmation": "reproduced"
    },
    {
      "test_id": "ROB-MONO-01",
      "severity": "high",
      "confirmation": "reproduced"
    },
    {
      "test_id": "DQ-CAT-01",
      "severity": "high",
      "confirmation": "reproduced"
    }
  ]
}
```

ModelAudit — PERF-AUC-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "auc": 0.7836948570757338,
    "auc_ci95": [
      0.7703127158591869,
      0.7981650080238237
    ],
    "ks": 0.4284824712626417,
    "gini": 0.5673897141514677,
    "count": 6000
  },
  "threshold": {
    "metric": "auc",
    "warn_below": 0.75,
    "fail_below": 0.7
  }
}
```

ModelAudit — CAL-ECE-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "brier": 0.1293283414401038,
    "ece": 0.02324224443776516,
    "calibration_in_the_large": -0.003090745277377932,
    "hosmer_lemeshow_stat": 29.891477235316067,
    "hosmer_lemeshow_df": 8,
    "hosmer_lemeshow_p": 0.00022092081386787028,
    "reliability_bins": [
      {
        "bin": 0,
        "count": 600,
        "observed_rate": 0.051666666666666666,
        "predicted_rate": 0.07324362398348597
      },
      {
        "bin": 1,
        "count": 600,
        "observed_rate": 0.06333333333333334,
        "predicted_rate": 0.0936312071641383
      },
      {
        "bin": 2,
        "count": 600,
        "observed_rate": 0.085,
        "predicted_rate": 0.10930947738704729
      },
      {
        "bin": 3,
        "count": 600,
        "observed_rate": 0.09833333333333333,
        "predicted_rate": 0.12381780763090343
      },
      {
        "bin": 4,
        "count": 600,
        "observed_rate": 0.10833333333333334,
        "predicted_rate": 0.13832949907680706
      },
      {
        "bin": 5,
        "count": 600,
        "observed_rate": 0.16333333333333333,
        "predicted_rate": 0.15533682664036366
      },
      {
        "bin": 6,
        "count": 600,
        "observed_rate": 0.20333333333333334,
        "predicted_rate": 0.1792204979389362
      },
      {
        "bin": 7,
        "count": 600,
        "observed_rate": 0.24333333333333335,
```

```
        "predicted_rate": 0.22725554028534817
      },
      {
        "bin": 8,
        "count": 600,
        "observed_rate": 0.39,
        "predicted_rate": 0.3708807057008865
      },
      {
        "bin": 9,
        "count": 600,
        "observed_rate": 0.7033333333333334,
        "predicted_rate": 0.6698822669658626
      }
    ]
  },
  "threshold": {
    "metric": "ece",
    "warn_above": 0.03,
    "fail_above": 0.05
  }
}
```

ModelAudit — STAB-PSI-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "psi": 0.009013948217124014,
    "baseline_test_psi": 0.012157855610367906,
    "scenario": "limit -20% and age +5"
  },
  "threshold": {
    "metric": "psi",
    "warn_above": 0.1,
    "fail_above": 0.25
  }
}
```

ModelAudit — STAB-CSI-01 (page 1)


```
{
  "status": "fail",
  "values": {
    "max_csi": 0.29407266523914694,
    "feature": "LIMIT_BAL",
    "baseline_test_csi": 0.012817605796263731
  },
  "threshold": {
    "metric": "max_csi",
    "warn_above": 0.1,
    "fail_above": 0.25
  }
}
```

ModelAudit — FAIR-SEX-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "threshold": 0.25,
    "groups": {
      "female": {
        "approval_rate": 0.7907053394858273,
        "good_outcome_tpr": 0.867563162184189
      },
      "male": {
        "approval_rate": 0.7670262980445044,
        "good_outcome_tpr": 0.8614035087719298
      }
    },
    "demographic_parity_difference": 0.023679041441322934,
    "equalized_odds_difference": 0.01216447170001006,
    "approval_rate_ratio": 0.9700532673059717,
    "overall_approval_rate": 0.779,
    "approved_observed_default_rate": 0.12430466409927257
  },
  "threshold": {
    "metric": "equalized_odds_difference",
    "warn_above": 0.05,
    "fail_above": 0.1
  }
}
```

ModelAudit — FAIR-AGE-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "threshold": 0.25,
    "groups": {
      "35plus": {
        "approval_rate": 0.7747234327857861,
        "good_outcome_tpr": 0.8615185504745471
      },
      "under35": {
        "approval_rate": 0.7832283725555187,
        "good_outcome_tpr": 0.8675496688741722
      }
    },
    "demographic_parity_difference": 0.008504939769732545,
    "equalized_odds_difference": 0.027920883740132352,
    "approval_rate_ratio": 0.9891411750802864,
    "overall_approval_rate": 0.779,
    "approved_observed_default_rate": 0.12430466409927257
  },
  "threshold": {
    "metric": "equalized_odds_difference",
    "warn_above": 0.05,
    "fail_above": 0.1
  }
}
```

ModelAudit — ROB-MONO-01 (page 1)


```
{
  "status": "fail",
  "values": {
    "test_feature": "PAY_0 missed-payment months",
    "violations": 66,
    "sample_rows": 400,
    "max_absolute_change": 0.2632604534713758,
    "missing_predictions_finite": true
  },
  "threshold": {
    "metric": "violations",
    "warn_above": 0,
    "fail_above": 1
  }
}
```

ModelAudit — DQ-CAT-01 (page 1)


```
{
  "status": "fail",
  "values": {
    "rows": 6000,
    "null_cells": 0,
    "duplicate_feature_rows": 2,
    "invalid_categories": 103,
    "label_out_of_range": 0,
    "id_and_target_excluded": true
  },
  "threshold": {
    "metric": "invalid_categories",
    "warn_above": 0,
    "fail_above": 100
  }
}
```

ModelAudit — FAIR-TRADEOFF (page 1)


```
{
  "status": "information",
  "values": {
    "thresholds": [
      {
        "threshold": 0.15,
        "groups": {
          "female": {
            "approval_rate": 0.5560316413974951,
            "good_outcome_tpr": 0.632436837815811
          },
          "male": {
            "approval_rate": 0.48988536749831424,
            "good_outcome_tpr": 0.5789473684210527
          }
        },
        "demographic_parity_difference": 0.06614627389918082,
        "equalized_odds_difference": 0.05348946939475829,
        "approval_rate_ratio": 0.8810386514462866,
        "overall_approval_rate": 0.5233333333333333,
        "approved_observed_default_rate": 0.08535031847133757
      },
      {
        "threshold": 0.2,
        "groups": {
          "female": {
            "approval_rate": 0.7336849044166117,
            "good_outcome_tpr": 0.8125509372453138
          },
          "male": {
            "approval_rate": 0.6894807821982468,
            "good_outcome_tpr": 0.787719298245614
          }
        },
        "demographic_parity_difference": 0.044204122218364894,
        "equalized_odds_difference": 0.03702623906705543,
        "approval_rate_ratio": 0.9397505360240256,
        "overall_approval_rate": 0.7118333333333333,
        "approved_observed_default_rate": 0.11261999531725592
      },
      {
        "threshold": 0.25,
        "groups": {
          "female": {
            "approval_rate": 0.7907053394858273,
            "good_outcome_tpr": 0.867563162184189
          },
          "male": {
            "approval_rate": 0.7670262980445044,
            "good_outcome_tpr": 0.8614035087719298
          }
        },
        "demographic_parity_difference": 0.023679041441322934,
        "equalized_odds_difference": 0.01216447170001006,
        "approval_rate_ratio": 0.9700532673059717,
        "overall_approval_rate": 0.779,
```

ModelAudit — FAIR-TRADEOFF (page 2)


```
        "approved_observed_default_rate": 0.12430466409927257
      },
      {
        "threshold": 0.3,
        "groups": {
          "female": {
            "approval_rate": 0.8226763348714569,
            "good_outcome_tpr": 0.8964955175224124
          },
          "male": {
            "approval_rate": 0.7963587322993931,
            "good_outcome_tpr": 0.8894736842105263
          }
        },
        "demographic_parity_difference": 0.026317602572063725,
        "equalized_odds_difference": 0.02346436111390371,
        "approval_rate_ratio": 0.9680097731555924,
        "overall_approval_rate": 0.8096666666666666,
        "approved_observed_default_rate": 0.12968299711815562
      },
      {
        "threshold": 0.35,
        "groups": {
          "female": {
            "approval_rate": 0.8599208965062624,
            "good_outcome_tpr": 0.9299103504482478
          },
          "male": {
            "approval_rate": 0.8256911665542819,
            "good_outcome_tpr": 0.9153508771929825
          }
        },
        "demographic_parity_difference": 0.03422972995198048,
        "equalized_odds_difference": 0.036096310445360325,
        "approval_rate_ratio": 0.9601943270700235,
        "overall_approval_rate": 0.843,
        "approved_observed_default_rate": 0.1362198497429814
      }
    ]
  },
  "threshold": null
}
```

ModelAudit — EXPL-SHAP-01 (page 1)


```
{
  "status": "information",
  "values": {
    "dataset": "taiwan",
    "rows_explained": 100,
    "xgb_shap_global_top10": [
      {
        "feature": "num__PAY_0",
        "mean_abs_shap": 0.5742000341415405
      },
      {
        "feature": "num__PAY_AMT2",
        "mean_abs_shap": 0.0914265364408493
      },
      {
        "feature": "num__PAY_5",
        "mean_abs_shap": 0.0797201469540596
      },
      {
        "feature": "num__PAY_AMT1",
        "mean_abs_shap": 0.07878690958023071
      },
      {
        "feature": "num__LIMIT_BAL",
        "mean_abs_shap": 0.07741788774728775
      },
      {
        "feature": "num__PAY_AMT3",
        "mean_abs_shap": 0.06772103905677795
      },
      {
        "feature": "num__PAY_2",
        "mean_abs_shap": 0.06638272851705551
      },
      {
        "feature": "num__BILL_AMT1",
        "mean_abs_shap": 0.06513115018606186
      },
      {
        "feature": "num__PAY_4",
        "mean_abs_shap": 0.05817516893148422
      },
      {
        "feature": "num__PAY_3",
        "mean_abs_shap": 0.05667481571435928
      }
    ],
    "xgb_first_case_adverse_top3": [
      {
        "feature": "num__PAY_0",
        "shap_log_odds": 1.6539591550827026
      },
      {
        "feature": "num__PAY_2",
        "shap_log_odds": 0.18914002180099487
      },
```

ModelAudit — EXPL-SHAP-01 (page 2)


```
      {
        "feature": "num__LIMIT_BAL",
        "shap_log_odds": 0.05639992654323578
      }
    ],
    "scorecard_first_case_reason_codes": [
      "PAY_0",
      "LIMIT_BAL",
      "PAY_AMT1"
    ],
    "scorecard_top3_overlap_under_1pct_amount_change": 0.9833333333333333,
    "warning": "SHAP is model explanation in log-odds space, not a causal reason or lawful adverse-action
  notice."
  },
  "threshold": null
}
```

ModelAudit — SURV-01 (page 1)


```
{
  "status": "information",
  "values": {
    "dataset": "UCI Taiwan; retrospective April\u2013September payment statuses",
    "people": 10000,
    "person_month_rows": 50973,
    "event_definition": "first recorded payment delay of at least two months",
    "events": 2801,
    "censored": 7199,
    "cox_concordance_train": 0.6520831166192114,
    "cox_hazard_ratios": {
      "LIMIT_BAL": 0.7486072184023408,
      "SEX": 0.9152345011329724,
      "AGE": 0.9504288600537097,
      "EDUCATION": 1.0427342693473656,
      "MARRIAGE": 0.9900729352017101
    },
    "ph_test_p": {
      "AGE": 0.8877453482639952,
      "EDUCATION": 0.04681646040104071,
      "LIMIT_BAL": 0.05282408133367339,
      "MARRIAGE": 0.9230135173119021,
      "SEX": 0.11013594798910528
    },
    "kaplan_meier": {
      "limit_below_100k": {
        "people": 3903,
        "survival_at_3": 0.7494235203689473,
        "survival_at_6": 0.5936459133999489
      },
      "limit_at_least_100k": {
        "people": 6097,
        "survival_at_3": 0.8807610300147616,
        "survival_at_6": 0.8007216663933082
      }
    },
    "discrete_hazard_train_auc": 0.7091065046785967,
    "warning": "Exploratory in-sample survival fit; event is delinquency proxy, not observed default. PH
  checks are diagnostics, not proof. Static default label is a different endpoint and later features can
  leak into early-event prediction."
  },
  "threshold": null
}
```